



# Dijkstra and Bellman–Ford

## 1 Dijkstra

**Exercise 1.** Give an example

```
1 for  $v \in V$  do
2    $d[v] \leftarrow \infty$ 
3    $\pi[v] \leftarrow \text{null}$ 
4 end
5  $d[s] \leftarrow 0$ 
6 for  $i \in [n]$  do
7   for  $(u, v) \in E$  do
8     if  $d[v] > d[u] + w(u, v)$  then
9        $d[v] \leftarrow d[u] + w(u, v)$ 
10       $\pi[v] \leftarrow u$ 
11     end
12   end
13 end
14 for  $(u, v) \in E$  do
15   if  $d[v] > d[u] + w(u, v)$  then
16     return Negative Cycle
17   end
18 end
```

**Algorithm 1:** Bellman–Ford.